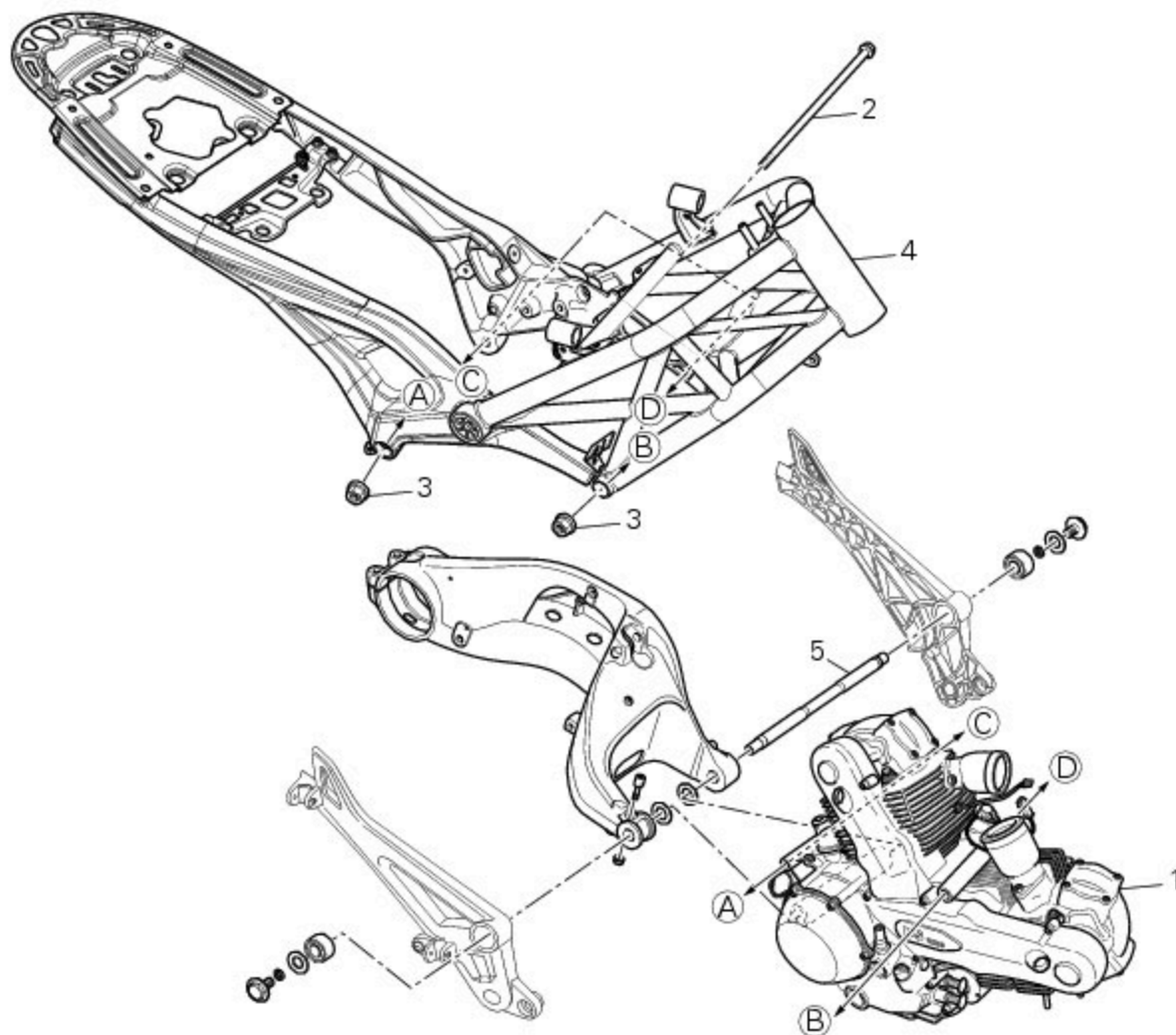


1 -Removal-refitting of the engine assembly



- 1 Engine
- 2 Bolt
- 3 Nut
- 4 Complete frame assembly
- 5 Swingarm pivot shaft

 Spare parts catalogue

1100 ABS	FRAME
1100 ABS	SWINGARM
1100 S ABS	FRAME
1100 S ABS	SWINGARM



Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Removal of the engine

In order to remove engine you must first remove a series of other components from the motorcycle.

Most of these removal procedures are described in the relative sections of this manual.

The following flow chart illustrates the logical sequence in which the parts are to be removed from the motorcycle and a reference to the section where the removal procedure is described.

This section describes only the operations to be carried out after having removed all the parts listed in the flow chart.

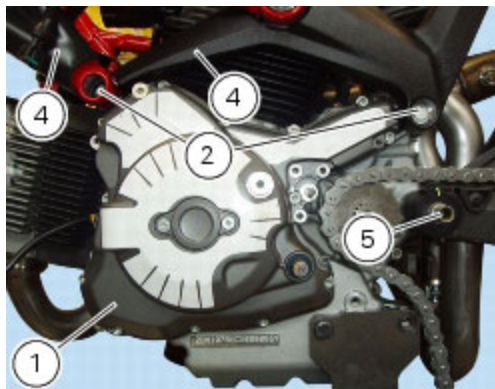
Operation	Section reference
Remove the seat	E 3, Removal of the seat
Remove the tank covers	E 2, Removal of the fuel tank fairings
Remove the fuel tank	L 2, Removal of the fuel tank
Remove the battery	P 2, Removal of the battery
Remove the battery mount	P 2, Battery support
Disconnect ECU connector from main wiring harness	M 3, Removal of the electronic control unit
Connect the starter motor/solenoid cable	P 3, Removal of the starter motor
Disconnect the HT leads	M 3, Spark plug
Refit the ABS control unit	G 9.2, Removing of the ABS control unit
Remove the airbox	L 7, Removal of the airbox
Disconnect the throttle cable	F 1, Removal of the throttle twistgrip
Remove the throttle body	L 6, Removal of the throttle body
Remove the oil cooler from the frame	N 2.2, Removal of the oil cooler
Remove the gearchange control	F 5, Removal of the gearchange control
Remove the rear brake fluid reservoir	F 4, Removal of the rear brake master cylinder
Remove the footrests	H 4, Removal of the footrest brackets
Remove the clutch slave cylinder	F 2, Removal of the clutch slave cylinder
Remove the front sprocket	G 8, Removing of the front sprocket
Remove the sidestand	H 5, Removing of the sidestand
Remove the vertical head exhaust pipe and the horizontal head exhaust pipe	L 8, Removal of the exhaust system
Disconnect wiring connectors on the engine block	P 1, Routing of wiring on frame

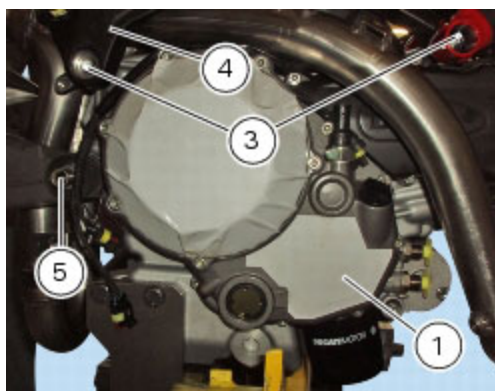
Place a stand under the engine.

Unscrew and remove the nuts (3) on the right-hand side of the frame in correspondence with front and rear mounting bolts (2) securing the engine (1).

Slide out swingarm pivot shaft (5) to remove the swinging arm from engine (1) (Sect. G 5, [Removal of the swingarm](#)).

Withdraw the complete engine block (1) from the frame (4).





Refitting the engine

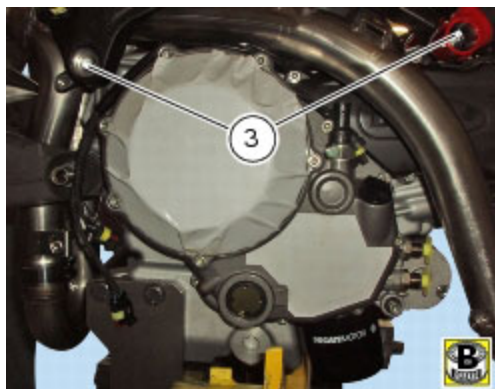
Refitting is the reverse of removal.

Important

Position the correct shims between engine and swingarm (Sect. G 5, [Refitting the swingarm](#)).

Lubricate the nuts (3) of mounting bolts (2).

Tighten the nuts (3) to the specified torque (Sect. C 3, [Engine torque settings](#)).



Operation	Section reference
Connect wiring connectors on the engine block	P 1, Routing of wiring on frame
Refit the vertical head exhaust pipe and the horizontal head exhaust pipe	L 8, Refitting the exhaust system
Refit the sidestand	H 5, Refitting the sidestand
Refit the front sprocket cover	G 8, Refitting the front sprocket
Refit the clutch slave cylinder	F 2, Refitting the clutch slave cylinder

Refit the footrests	H 4, Refitting the footrest brackets
Refit the rear brake fluid reservoir	F 4, Refitting the rear brake master cylinder
Refit the gearchange control	F 5, Refitting the gearchange mechanism
Refit the oil cooler to the frame	N 2.2, Refitting the oil cooler
Refit the throttle body	L 6, Refitting the throttle body
Connect the throttle cable	F 1, Refitting the throttle twistgrip
Refit the airbox	L 7, Refitting the airbox
Reconnect the HT leads	M 3, Spark plug
Refit the ABS control unit	G 9.2, Refitting the ABS control unit
Connect the starter motor/solenoid cable	P 3, Refitting the starter motor
Connect ECU connector to main wiring harness	M 3, Refitting the electronic control unit
Refit the battery mount	P 2, Battery support
Refit the battery	P 2, Refitting the battery
Refit the fuel tank	L 2, Refitting the fuel tank
Refit the tank covers	E 2, Refitting the fuel tank fairings
Refit the seat	E 3, Refitting the seat